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GitHub Link	https://github.com/kiduuuu/MLA0405-Deep-learning/tree/main

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

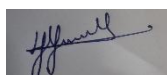
Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I D.Chinna Moulali/ 192425278 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05**Total Marks:25****Q1. CNN vs Fully Connected Neural Network**

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Aspect	CNN	Fully Connected Neural Network
Architecture	Uses convolution, pooling and fully connected layers	Mainly consists of fully connected layers
Motivation	Designed specifically to process spatial data such as images	Designed for general numerical/vector data
Spatial features	Preserves spatial relationships between nearby pixels	Loses spatial relationships after flattening
Parameters	Fewer due to parameter sharing	Very large because every neuron connects to every input
Computation	More efficient for images	Computationally expensive for high-resolution images
Image handling	Highly suitable	Less suitable
Feature extraction	Automatically learns edges, textures and shapes	Requires a large number of weights to learn image patterns

Role of CNN Layers

- **Convolutional layers:** Extract local image features using filters.
- **Filters:** Small matrices that slide across an image and detect patterns such as edges, corners and textures.
- **Pooling layers:** Reduce feature-map dimensions and computation while retaining important information.
- **Fully connected layers:** Use the extracted features for final classification.

For example, an X-ray may initially contain thousands of pixels. Connecting every pixel to every neuron in a fully connected network creates millions of parameters. A CNN instead uses small filters repeatedly across the image.

Conclusion: CNNs are more suitable for X-ray classification because they preserve spatial information, require fewer parameters through parameter sharing, automatically learn visual features and provide better computational efficiency.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Layer	Main Function	Spatial Information	Computational Effect
Convolution	Extracts visual features	Preserves local relationships	Moderate computation
Pooling	Reduces feature-map size	Retains important features	Reduces computation
Activation	Adds non-linearity	Maintains feature-map structure	Low additional cost
Fully Connected	Performs final classification	Uses extracted features	Can contain many parameters

- **Convolutional Layer**

Convolutional layers apply filters to the product image. They can identify scratches, cracks, holes, irregular edges and other defect patterns.

- **Pooling Layer**

Pooling, such as **max pooling**, reduces the spatial dimensions of feature maps. This makes the model faster and less sensitive to small changes in defect location.

- **Activation Layer**

ReLU is commonly used. It introduces non-linearity, allowing the CNN to learn complex defect patterns.

- **Fully Connected Layer**

The final learned features are combined by fully connected layers to classify the product as **defective or non-defective**.

- **Early vs Deeper Filters**

Early-layer filters: Learn simple features such as edges, lines, corners and basic textures.

- **Middle-layer filters:** Combine simple features to learn shapes and surface patterns.

- **Deep-layer filters:** Learn complex patterns representing complete defects or combinations of defect features.

Conclusion: CNN layers work hierarchically. Early layers detect basic visual information, while deeper layers combine these features to identify complex manufacturing defects accurately.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Aspect	CNN Parameter Sharing	Fully Connected Independent Weights
Weights	Same filter reused across image	Different weight for each connection
Parameters	Much fewer	Very large
Memory	Lower	Higher
Computation	More efficient for images	Expensive for high-resolution images
Feature detection	Same feature can be detected anywhere	Location-specific connections
Translation handling	Good ability to detect shifted features	Less efficient
Learning efficiency	High	Lower for image data

In a CNN, a **3×3 filter contains only 9 weights plus a bias** and these weights are reused at different image locations. Thus, the network does not need separate weights for every location.

For example, a filter trained to detect a vertical edge can detect that edge whether it occurs on the left, center or right side of an image.

Advantages

1. **Fewer parameters** → less memory usage.
2. **Lower computational requirements** → faster processing.
3. **Better learning efficiency** → the same useful feature is learned from many locations.
4. **Translation-related feature detection** → objects can be recognized even when their position changes.
5. **Better scalability** → particularly useful for thousands of high-resolution surveillance images.

Conclusion: Parameter sharing is one of the main reasons CNNs are highly effective for large-scale image processing.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature

extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Feature	AlexNet	ResNet
Depth	8 learned layers	18, 34, 50, 101+ layers
Architecture	Conventional sequential CNN	Deep CNN with residual/skip connections
Filters	Convolutional filters	Multiple convolutional filters in deeper blocks
Feature extraction	Good	More powerful and hierarchical
Parameters	Relatively high for its depth	Varies by version; deeper models can be parameter-efficient for their capability
Training	Relatively simpler	Easier to train deep networks because of skip connections
Vanishing gradient	More susceptible	Significantly reduced
Computational cost	Generally lower	Higher for larger versions
Classification performance	Good	Generally better on complex recognition tasks

AlexNet

AlexNet is an early deep CNN architecture containing convolutional, pooling and fully connected layers. It demonstrated the effectiveness of deep learning for image recognition.

ResNet

ResNet introduces **residual connections**, which allow information and gradients to bypass some layers. This enables much deeper networks to be trained effectively.

Recommendation

ResNet is more suitable for complex image recognition.

Reasons:

- Learns more complex hierarchical features.
- Handles deeper architectures effectively.
- Reduces the vanishing-gradient problem.
- Usually provides better classification performance.
- Supports transfer learning from pretrained models.

For applications with limited computational resources, **ResNet-18 or ResNet-34** can provide a good balance between accuracy and computational cost.

Conclusion: AlexNet is simpler and computationally lighter, but ResNet is generally preferred for modern complex image-recognition applications because of its superior feature learning and deep-network training capability.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Aspect	CNN Without Regularization	CNN With Regularization
Training accuracy	Often very high	May be slightly lower
Overfitting	High	Reduced
Generalization	Poor	Better
Training stability	Can be less stable	Batch normalization improves stability
Unseen images	Poor performance	Better performance
Data requirement	Needs more data	Data augmentation increases variety
Reliability	Lower	Higher

The initial agricultural CNN has **high training accuracy but poor unseen-image performance**, which is a clear indication of **overfitting**.

1. Data Augmentation

Training images can be randomly:

- Rotated
- Flipped where medically/biologically appropriate
- Cropped
- Scaled
- Adjusted in brightness and contrast

This creates more variation and helps the CNN recognize plant diseases under different real-world conditions.

2. Dropout

Dropout randomly disables some neurons during training. It prevents the network from becoming overly dependent on particular features and improves generalization.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training. It:

- Improves training stability.
- Allows faster and smoother training.
- Can provide a regularization effect.
- Helps the network converge more effectively.

Recommended Approach

The agricultural organization should use a **regularized CNN combining data augmentation, dropout and batch normalization**.

A suitable structure is:

Input → Convolution → Batch Normalization → ReLU → Pooling → Convolution → Batch Normalization → ReLU → Pooling → Dropout → Fully Connected → Output

Data augmentation is particularly important because real-world leaf images can differ in **lighting, orientation, size and background**. Dropout helps prevent feature dependency, while batch normalization improves training stability.

Conclusion: A regularized CNN will generalize much better to unseen plant images and is therefore more reliable for real-world plant disease classification than a non-regularized CNN.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and	Some insights derived from comparison	Limited or unclear insights	No insights generated

		conclusions from comparisons			
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand